# Block practice · More bones, muscles and joints

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Draw or label the parts of a synovial joint

Explain dietary needs for bone and muscle and describe osteoporosis

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# Practice workbook · Draw or label the parts of a synovial joint

Internal structure of a synovial joint shown as a diagram.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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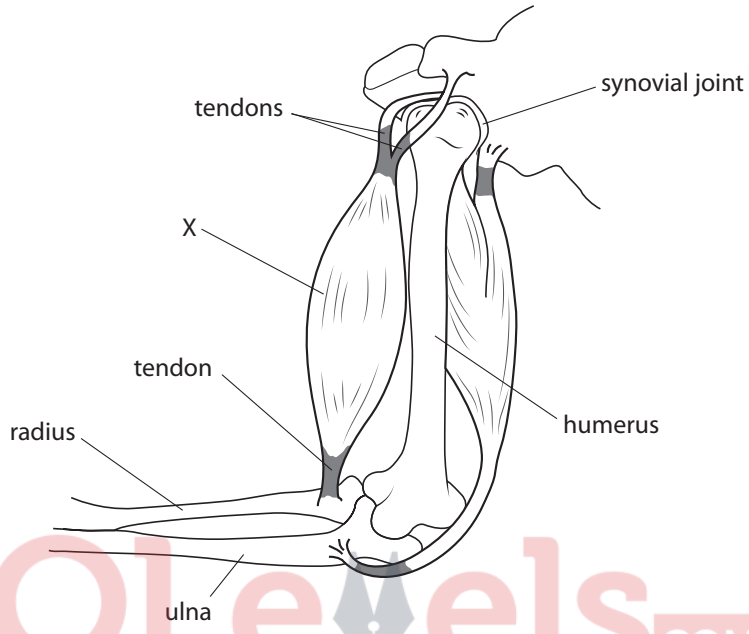
Type ID: HBI-SYNOVIAL-JOINT-STRUCTURE

# 1. Question 7 · part b · 6 marks

November 2020 Paper 1

Worked-example retry · Attempt part b only.

**7** The diagram shows the bones of a human arm and the muscles that move the lower arm.




(a) (i) Muscle X needs energy to contract.

Name the process that supplies this energy.

**(1)**

(ii) Muscle X contracts when the lower arm is raised further.

Describe the changes in shape of muscle X when the lower arm is raised further.

**(2)**

**20**

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# 1. Question 7 · part b · 6 marks

November 2020 Paper 1

Worked-example retry · Attempt part b only.

(iii) Explain how the contraction of muscle X causes the lower arm to be raised.

**(3)**

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(b) Draw a labelled diagram to show the internal structure of the synovial joint.

**(6)**

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**(Total for Question 7 = 12 marks)**

## 1. Question 7 · part b · 6 marks · Working and answer

Working and answer

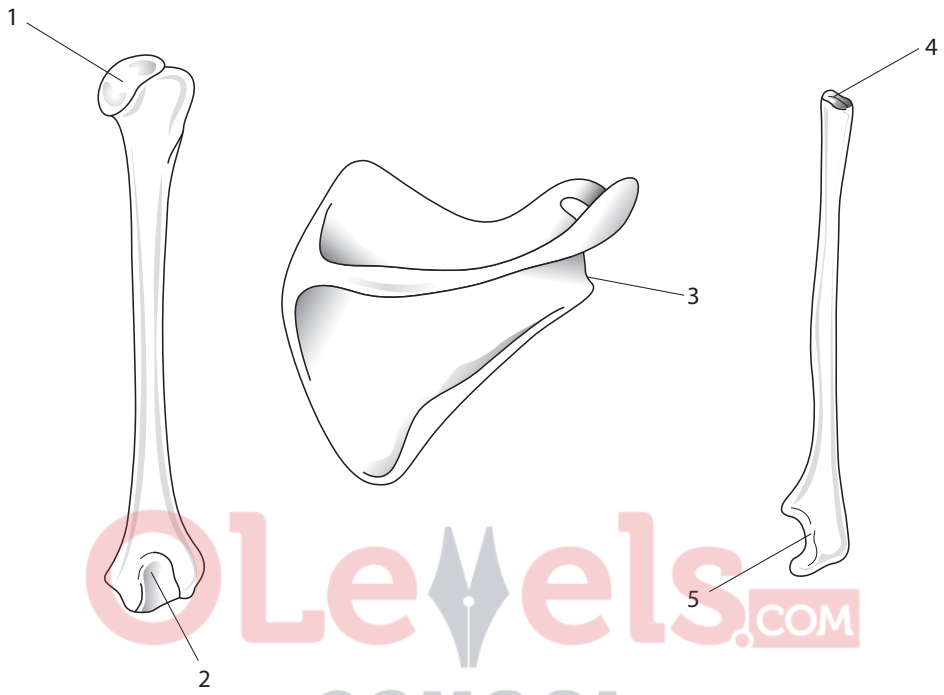
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# 2. Question 2 · part b.iii · 1 mark

June 2019 Paper 2R

Further practice · Attempt part b.iii only.

**2** (a) The diagrams show three bones from the human skeleton.




Which two numbered regions come together to form a ball and socket joint?

**(1)**

- **A** 1 and 3
- **B** 1 and 5
- **C** 2 and 3
- **D** 2 and 4

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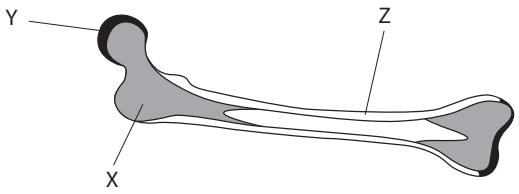
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# 2. Question 2 · part b.iii · 1 mark

June 2019 Paper 2R

Further practice · Attempt part b.iii only.

(b) The diagram shows a section through a bone.




(i) Identify the part labelled X.

(1)

- ☒ **A** cartilage
- ☒ **B** compact bone
- ☒ **C** ligament
- ☒ **D** spongy bone

(ii) Identify the part labelled Z.

(1)

- ☒ **A** cartilage
- ☒ **B** compact bone
- ☒ **C** ligament
- ☒ **D** spongy bone

(iii) Name the part labelled Y.

(1)

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(iv) Describe the function of the part labelled Y.

(2)

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## 2. Question 2 · part b.iii · 1 mark · Working and answer

Working and answer

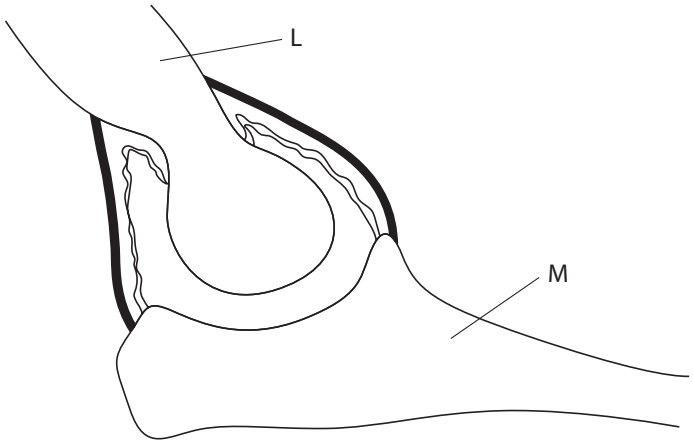
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## 3. Question 6 · part b.ii · 2 marks

November 2021 Paper 2

Further practice · Attempt part b.ii only.

**6** The diagram shows some of the structures of the elbow joint.




(a) State what is meant by the term **joint**.

**(2)**

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(b) (i) Give the name of bone L and the name of bone M.

**(2)**

L .....

M .....

(ii) Cartilage has not been shown on the diagram.

Draw the missing cartilage on the diagram.

**(2)**

**20**

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## 3. Question 6 · part b.ii · 2 marks

November 2021 Paper 2

Further practice · Attempt part b.ii only.

(c) Explain the effects on a person if the cartilage in this joint is missing or damaged.

**(2)**

(d) Tendons are important in the functioning of a joint.

Explain the importance of tendons in the movement of bone M.

**(4)**

**(Total for Question 6 = 12 marks)**

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## 3. Question 6 · part b.ii · 2 marks · Working and answer

Working and answer

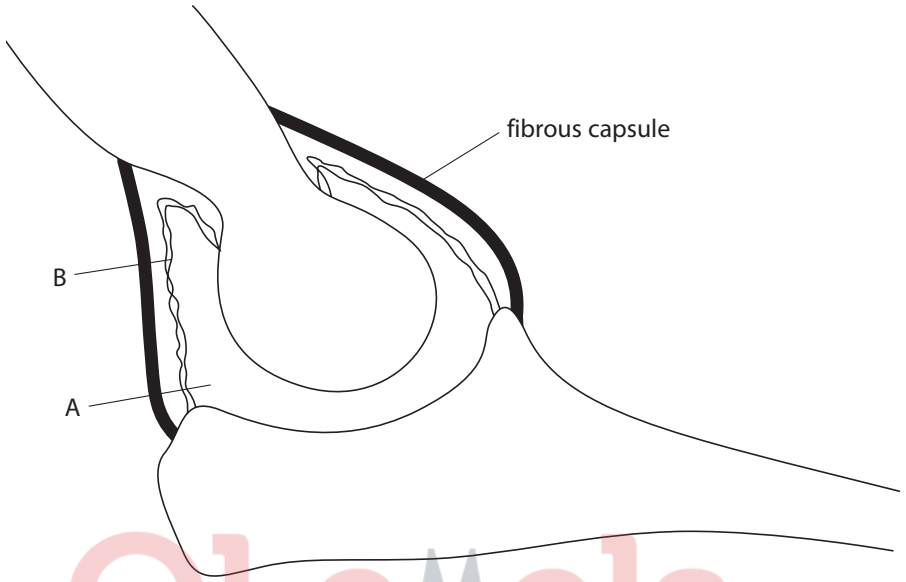
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# 4. Question 3 · part a.i · 2 marks

January 2022 Paper 1

Further practice · Attempt part a.i only.

**3** The diagram shows part of an elbow joint without any cartilage.




(a) (i) Identify part A and part B. **(2)**

part A .....

part B .....

(ii) Describe the function of part A. **(2)**

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(b) (i) Complete the diagram by drawing the cartilage. **(2)**

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## 4. Question 3 · part a.i · 2 marks · Working and answer

Working and answer

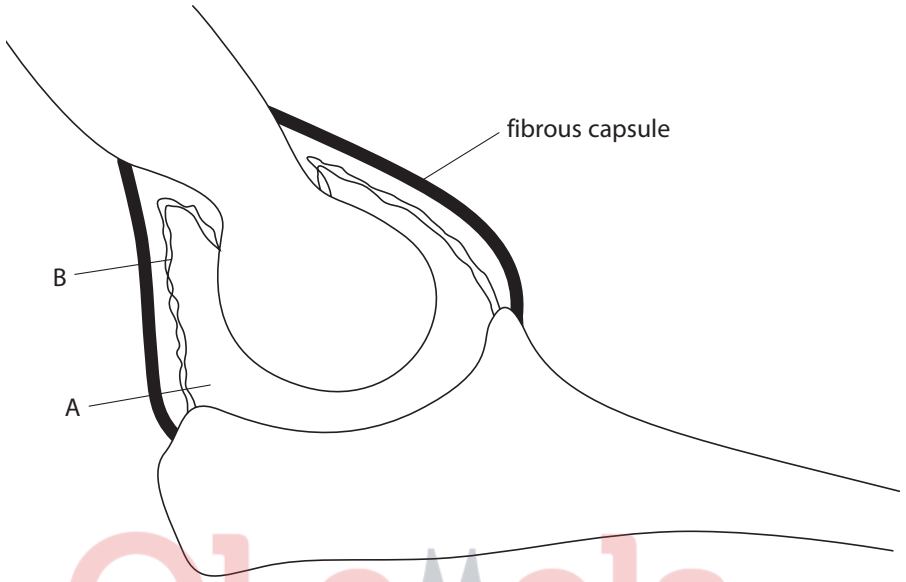
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## 5. Question 3 · part b.i · 2 marks

January 2022 Paper 1

Further practice · Attempt part b.i only.

**3** The diagram shows part of an elbow joint without any cartilage.




(a) (i) Identify part A and part B. **(2)**

part A .....

part B .....

(ii) Describe the function of part A. **(2)**

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(b) (i) Complete the diagram by drawing the cartilage. **(2)**

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## 5. Question 3 · part b.i · 2 marks

January 2022 Paper 1

Further practice · Attempt part b.i only.

(ii) Cartilage often breaks down as people become older.

Explain the effects this has on the working of the joint.

**(2)**

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(c) Explain the role of tendons in the movement of the lower arm.

**(5)**

**(Total for Question 3 = 13 marks)**

## 5. Question 3 · part b.i · 2 marks · Working and answer

Working and answer

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Practice workbook · Explain dietary needs for bone and muscle and describe osteoporosis

Calcium, vitamin D, phosphate and protein for growing and keeping bone and muscle, and the structure and symptoms of osteoporotic bone.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-BONE-HEALTH-DIET-AND-OSTEOPOROSIS

# 1. Question 2 · part d · 5 marks

June 2023 Paper 2

Worked-example retry · Attempt part d only.

**2** The table gives the nutritional information for a pot of yoghurt.

| | Mass per 100 g of yoghurt | Mass per 85 g pot |
|---|---|---|
| fat | 2.7 g | 2.3 g |
| carbohydrate | 12.6 g | 10.7 g |
| protein | 6.1 g | |
| calcium | 149 mg | 127 mg |
| vitamin D | 0.9 µg | 0.8 µg |

(a) Calculate the mass of protein in one pot of yoghurt.

Give your answer to two significant figures.

**(3)**

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mass of protein = ..... g

(b) Describe how the yoghurt could be tested to show that protein is present.

**(3)**

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# 1. Question 2 · part d · 5 marks

June 2023 Paper 2

Worked-example retry · Attempt part d only.

(c) A pot of yoghurt provides 16% of a person's daily requirement of calcium.

Calculate how many pots of yoghurt a person would need to eat to obtain their daily requirement of calcium.

**(2)**

number of pots = .....

(d) Explain why yoghurt is good for bone development.

**(5)**

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**(Total for Question 2 = 13 marks)**

## 1. Question 2 · part d · 5 marks · Working and answer

Working and answer

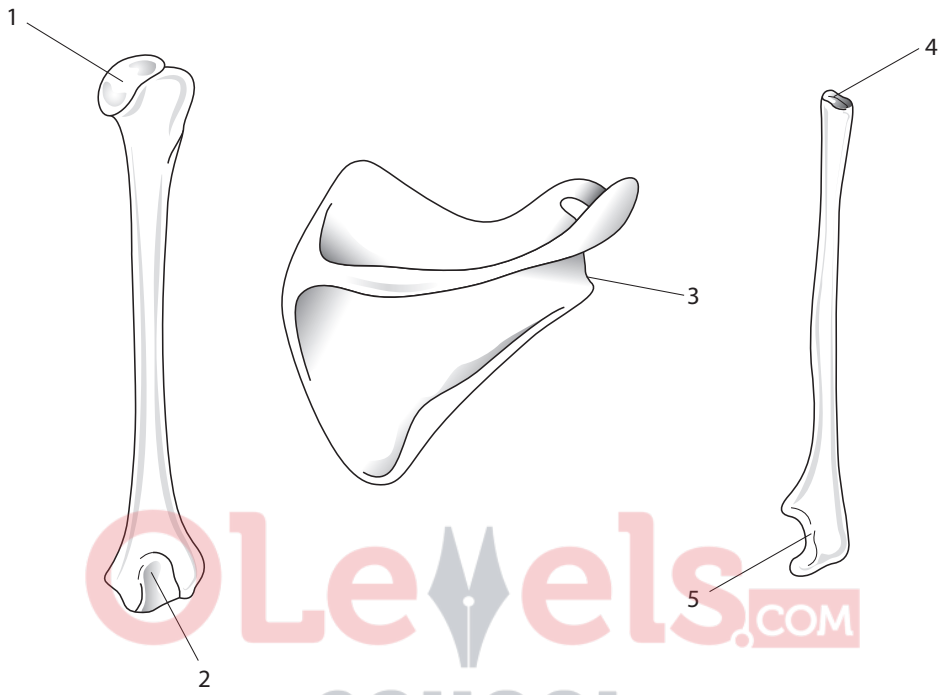
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## 2. Question 2 · part c · 6 marks

June 2019 Paper 2R

Further practice · Attempt part c only.

**2** (a) The diagrams show three bones from the human skeleton.




Which two numbered regions come together to form a ball and socket joint?

**(1)**

- ☒ **A** 1 and 3
- ☒ **B** 1 and 5
- ☒ **C** 2 and 3
- ☒ **D** 2 and 4

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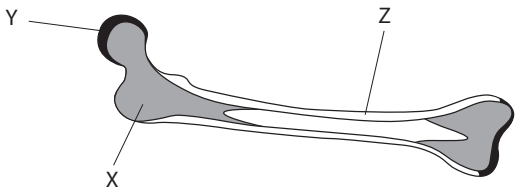
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## 2. Question 2 · part c · 6 marks

June 2019 Paper 2R

Further practice · Attempt part c only.

(b) The diagram shows a section through a bone.




(i) Identify the part labelled X.

**(1)**

- ☒ **A** cartilage
- ☒ **B** compact bone
- ☒ **C** ligament
- ☒ **D** spongy bone

(ii) Identify the part labelled Z.

**(1)**

- ☒ **A** cartilage
- ☒ **B** compact bone
- ☒ **C** ligament
- ☒ **D** spongy bone

(iii) Name the part labelled Y.

**(1)**

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(iv) Describe the function of the part labelled Y.

**(2)**

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## 2. Question 2 · part c · 6 marks

June 2019 Paper 2R

Further practice · Attempt part c only.

(c) Describe why a balanced diet is needed for the healthy development of bone and muscle.

**(6)**

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**(Total for Question 2 = 12 marks)**

## 2. Question 2 · part c · 6 marks · Working and answer

Working and answer

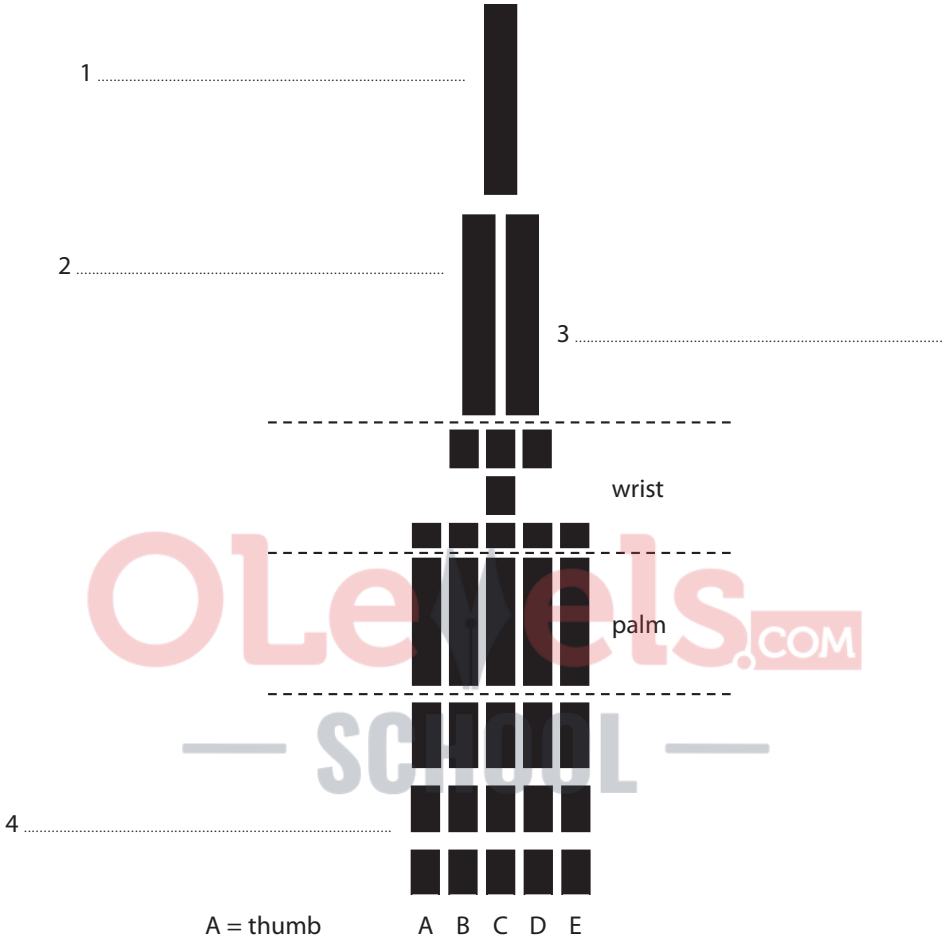
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## 3. Question 2 · part b · 3 marks

June 2021 Paper 1

Further practice · Attempt part b only.

**2** The diagram shows a model representing the arrangement of bones in the arm.




(a) (i) Name the bones labelled 1, 2, 3 and 4 on the diagram.

**(4)**

(ii) A hinge joint is a synovial joint.

Draw a circle on the diagram to show where a hinge joint is found.

**(1)**

(iii) Name two other types of synovial joint found in the skeleton.

**(2)**

1 .....

2 .....

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P 6 6 4 3 2 A 0 4 2 4

## 3. Question 2 · part b · 3 marks

June 2021 Paper 1

Further practice · Attempt part b only.

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(b) Vitamin D is important for the growth and development of healthy bones.

Explain how vitamin D helps the growth and development of bones.

**(3)**

**(Total for Question 2 = 10 marks)**

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Working and answer

# 4. Question 10 · part a · 3 marks

June 2022 Paper 1R

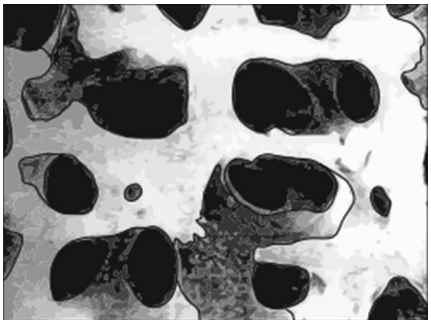
Further practice · Attempt part a only.

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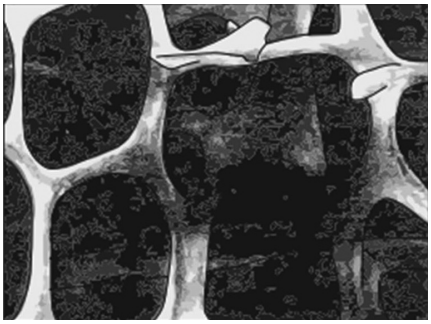
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**10** (a) The photographs show the structure of healthy bone and bone that is damaged by osteoporosis.



healthy bone



bone damaged by osteoporosis

Describe how osteoporosis might affect an individual.

**(3)**

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(b) Explain how voluntary muscles and bones bring about movement in the elbow joint.

**(3)**

**(Total for Question 10 = 6 marks)**

## 4. Question 10 · part a · 3 marks · Working and answer

Working and answer

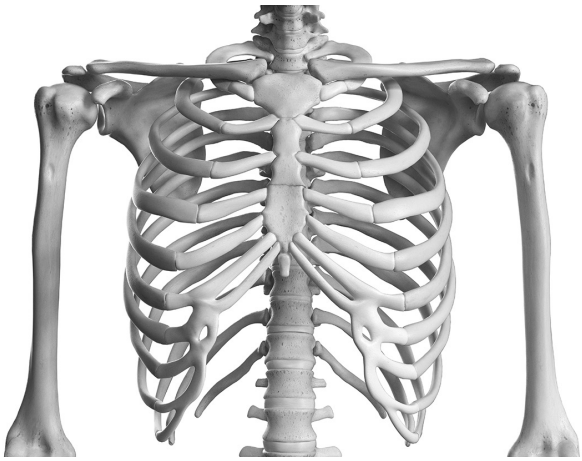
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## 5. Question 1 · part b · 3 marks

January 2023 Paper 2

Further practice · Attempt part b only.

**1** (a) The image shows part of the skeleton.



(Source: © SciePro/Shutterstock)

(i) Label the clavicle, scapula and vertebral column on the image.

**(3)**

(ii) The vertebral column is part of the axial skeleton.

Describe the functions of the axial skeleton.

**(2)**

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## 5. Question 1 · part b · 3 marks

January 2023 Paper 2

Further practice · Attempt part b only.

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(b) Osteoporosis can affect the vertebral column.

Describe the symptoms of osteoporosis.

**(3)**

**(Total for Question 1 = 8 marks)**

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Working and answer